Sahaj Bhadja

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Summary

Backend-focused engineer building **scalable APIs, business workflows, relational systems, automated tests, and cloud deployments**, combining C#/.NET project work with production Python, Azure, and SQL delivery.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Web Technologies, Analysis of Algorithms, Software Engineering

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Data Structures, Operating Systems, Computer Networks

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Architected a multi-tenant platform on **Azure SQL and Blob Storage** with normalized schemas, row-level isolation, and indexed reporting queries, reducing dashboard latency from 1.2 seconds to 350 milliseconds.
- Reworked asynchronous FastAPI workflows with **explicit job states, validation, failure isolation, logging, and customer support diagnostics**, tripling throughput for 1K+ monthly jobs while maintaining 99%+ production reliability.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Delivered React and Flask workflows for uploads, grading history, model controls, and feedback, supporting 150+ instructors and 10K+ daily requests through **end-to-end development, validation, testing, and deployment**.
- Implemented concurrent Python ingestion with MySQL and AWS RDS, reducing turnaround 60% through **secure file validation, resilient processing, integration tests, production debugging, and repeatable CI/CD-backed production releases**.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Designed object-oriented services and database integrations across Python, AWS Glue, S3, and SQL, improving response times 30% through **modular APIs, asynchronous processing, query optimization, and reusable domain interfaces**.
- Automated 20+ unit and integration suites with PyTest, Mockito, Jenkins, and virtual machines, increasing pipeline efficiency 68% through **repeatable builds, regression isolation, constructive reviews, and release validation**.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Built eight business services for authentication, billing, permissions, resources, and notifications, reliably synchronizing 2M+ records through **Node.js APIs, Azure SQL, WebSockets, event-driven updates, and shared relational models**.
- Developed React and TypeScript interfaces for operational records and real-time updates while automating Docker and Jenkins delivery, reducing deployment time 50% through **testing, monitoring, and reusable components**.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built Flask APIs for bookings, pricing, coupons, payments, and real-time status updates across 50K+ records, achieving sub-200-millisecond responses through **data modeling, webhook processing, caching, and query optimization**.
- Translated customer requirements into technical specifications for 12+ integrations, mapping **business workflows, validation rules, API contracts, data ownership, process impacts, and implementation milestones** before development and stakeholder review.

Skills

Languages & Frameworks: C#, Python, Java, TypeScript, .NET, ASP.NET Core, Web API, React, FastAPI
Backend: OOP, Async/Await, LINQ, REST APIs, Microservices, Workflows, Authentication, Validation, Error Handling, Scalability
Data: EF Core, SQL Server, Azure SQL, PostgreSQL, MySQL, SQL, Database Design, Data Modeling, Query Optimization
Cloud & Quality: Azure, AWS, Docker, Jenkins, CI/CD, Git, Logging, xUnit, NUnit, Unit Testing, Integration Testing, Debugging

Projects

Construction Project Controls Hub | *C#, ASP.NET Core, EF Core, SQL Server, Azure, React*

- Built a **project-controls application** for budgets, schedules, contracts, invoices, RFIs, and change orders, exposing role-secured REST APIs with EF Core transactions, validation, audit history, and operational dashboards.

Labor and Cost Workflow Service | *C#, .NET, SQL Server, Azure Functions, Docker, xUnit*

- Developed an **asynchronous labor and cost workflow engine** that validates time entries, allocates project costs, manages approval states, produces financial summaries, and verifies API behavior through xUnit integration tests.

Chat2SQL Code | *Java, TypeScript, SQL, Docker, Maven, Node.js* | GitHub

- Extended a cross-platform database workspace supporting 10+ databases through **object-oriented architecture, schema synchronization, validated query execution, containerized environments, and reusable integrations**, streamlining complex database development across heterogeneous systems.